

Beyond Representation: Index Structure Limitations in Dense Retrieval

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Abstract—Dense retrieval relies on approximate nearest-neighbor (ANN) indices such as HNSW and IVF for practical deployment. Prior theoretical work has focused on the *representational* capacity of dense embeddings; we show that the *indexing* stage imposes a separate, quantifiable bottleneck. We introduce *indexing bias*—the systematic gap between exact and approximate retrieval probability—and derive a centroid-dilution bound linking IDF-weighted token rarity, embedding dimensionality, and cluster granularity. We isolate index-induced failures empirically by holding the representation fixed (random-projection trigram embeddings at varying dimensions) and varying only the indexing structure on Home Depot product search and MS MARCO Passage; we further *directly measure* the centroid signal on each query, observing a $\sim 40\times$ collapse from standard to identifier queries that closes the inference gap between theory and observation. On so-called *spear-fishing* queries—those depending on identifiers or rare terms—switching from exact to HNSW search forfeits up to 63% of the achievable Recall@10, versus less than 2% for standard semantic queries. Increasing embedding dimensionality reduces representation noise but does not close this index-induced gap. The results reframe the sparse-vs-dense trade-off as a joint choice of representation and indexing, with direct implications for production retrieval stacks where part-number, SKU, and named-entity queries are operationally important.

Index Terms—dense retrieval, approximate nearest neighbor search, index structures, information retrieval, service quality

I. INTRODUCTION

Information retrieval (IR) has seen a paradigm shift with the advent of neural dense retrieval methods that use learned embeddings [1], challenging traditional sparse retrieval methods based on lexical term matching [2], [3]. As IR systems increasingly serve as critical components in various service ecosystems—from e-commerce platforms and enterprise search to digital assistants and knowledge management services—their reliability and performance across diverse query patterns directly impacts user satisfaction and service quality.

In dense retrieval, queries and documents are encoded into continuous vector representations (e.g., via BERT-based encoders) and relevant items are found by nearest-neighbor search in embedding space. This approach has revolutionized many information services by improving semantic understanding and results relevance. In contrast, sparse retrieval represents text as high-dimensional sparse vectors (often TF-IDF or BM25 term weights) and relies on inverted indexes to match overlapping terms, with recent advances also incorporating learned sparse representations [4].

Theoretical analyses, notably the work by Tay et al. [5] using random projection techniques, suggest that dense representations, given sufficient dimensionality, possess the theoretical capacity to approximate or even surpass traditional sparse methods. However, the practical scalability of dense retrieval in service environments hinges critically on Approximate Nearest Neighbor (ANN) search algorithms (e.g., HNSW, IVF). While much research has focused on the representational power of embeddings, the fundamental limitations imposed by these necessary ANN index structures themselves remain relatively underexplored. This gap becomes particularly concerning for service engineers designing systems that must handle heterogeneous query patterns from diverse user populations.

This paper argues that ANN index structures introduce significant bottlenecks, particularly for what we term “spear fishing” queries—those whose relevance depends critically on rare terms or specific, non-semantic patterns (like identifiers). These queries often correspond to documents lying in sparser regions of the embedding space but may represent important user service interactions such as precise product searches or specific information lookup tasks. We extend prior theoretical work [5], [6] by analyzing how the approximations inherent in common ANN indices systematically disadvantage such queries.

We make the following contributions:

- We formalize the concept of “spear fishing” queries and demonstrate theoretically why standard ANN indices (IVF and HNSW) struggle with them, *independent of the representation’s theoretical capacity*.
- We derive a *centroid-dilution bound* that links IDF-weighted token rarity, embedding dimensionality d , and clustering granularity to the probability that an IVF centroid fails to carry a rare token’s signal above the Johnson–Lindenstrauss noise floor, and we *directly measure* this signal on each query, showing it drops $\sim 40\times$ from standard to ID queries on Home Depot.
- We isolate index-induced failure empirically by comparing exact (Flat) and approximate (IVF, HNSW) search on a controlled three-axis design (two datasets $\times$ three query types $\times$ three embedding dimensions), and quantify the resulting *indexing bias*.
- We show that the gap is disproportionately large for ID and rare-term queries and that, unlike representation-level limitations, this gap does not vanish at higher embedding dimensionality.

What is new versus prior work. Tay et al. [5] and Fan et al. [6] analyze *representation-level* capacity of dense vectors. We complement this with an *index-level* analysis: we hold the representation fixed (via random-projection trigram embeddings of varying dimension) and vary only the indexing structure, so that every observed degradation can be attributed to the index. Sec. III formalizes this separation; Sec. IV tests it.

II. BACKGROUND AND RELATED WORK

A. Dense Retrieval Approaches

Dense retrieval models map queries and documents to continuous vector representations, typically using neural networks. Notable approaches include the Dense Passage Retriever (DPR) [2], which uses separate BERT encoders for queries and documents, ANCE [3], which employs hard negative mining to improve representation learning, and Sentence-BERT [8], which adapts BERT for efficient semantic similarity comparison. These models are trained on large datasets with relevance labels, learning to place semantically related queries and documents close in embedding space.

More recent work has explored multi-vector representations like ColBERT [9], which retains token-level representations to enable more fine-grained matching. While effective, these approaches generally require significant training data and computational resources.

B. Theoretical Capabilities of Dense Representations

Tay et al. [5] demonstrated that dense representations can theoretically approximate sparse retrieval performance using random projections. Their analysis showed that with sufficient dimensionality, dense vectors can capture the same information as sparse vectors. They introduced a latent-based dual encoder model that combines these theoretical insights with modern neural architectures, and proposed multi-vector representations to address some limitations of single-vector approaches.

Fan et al. [6] conducted an information-theoretic analysis of dense representations to measure how much variability in queries is preserved, finding that dense models have lower capacity for tail information than sparse ones. This theoretical limitation helps explain empirical observations about dense retrievers struggling with rare entities or terms.

C. Index Structures for Dense Retrieval

To make dense retrieval practical at scale, approximate nearest neighbor (ANN) indexing is essential [10]. Two families dominate in practice and frame our analysis: *Hierarchical Navigable Small World* (HNSW) graphs [11], which build a multi-layer proximity graph and search it greedily, and *Inverted File* (IVF) indices, optionally combined with Product Quantization [12], which cluster documents by k-means and probe only the closest centroids at query time. Both achieve speed via approximations that work well on average but can systematically fail on tail cases. Lin [7] notes that HNSW degrades retrieval quality by a small (often-unreported) amount

relative to exact search; we argue this small *average* loss masks a much larger loss concentrated on a specific query population.

D. Research Gap

Benchmarks such as BEIR [13] document dense-retrieval failures on out-of-distribution and entity-heavy queries, and fairness work [14] hints that index-level effects are involved, but the contribution of the ANN index itself is rarely isolated. The interaction between representation quality and index approximation quality remains open: even with embeddings of sufficient capacity, ANN search may introduce unavoidable per-query failure modes. This paper isolates the index contribution by construction.

III. THEORETICAL ANALYSIS

We extend the random-projection view of dense retrieval [5] to the indexing stage. The argument has three steps: (i) a representation has a token-level signal whose magnitude depends on IDF weight and dimensionality; (ii) exact search preserves this signal up to JL distortion; (iii) practical ANN structures introduce a *second*, qualitatively different distortion that disproportionately attenuates rare-token signals. Our contributions in this section are the centroid-dilution bound (Eq. (1)), the indexing-bias decomposition (Eq. (3)), and the four testable predictions in Sec. III-C that the experiments target directly.

A. Setup and Signal Magnitude

Following Tay et al. [5], we represent each document d as the IDF-weighted sum of its token embeddings $f(d) = \sum_{t \in d} \alpha_t \phi(t)$, with $\|\phi(t)\| = 1$ and $\alpha_t = \log(M/n_t)$ for corpus size M and document frequency n_t . High-dimensional embeddings (V buckets) are projected to dimension d by a random matrix W ; the Johnson–Lindenstrauss lemma [15] preserves inner products up to multiplicative distortion $\epsilon \approx \sqrt{\log V/d}$.

a) Centroid signal for a rare token.: Consider a cluster C of size $|C| \approx M/N$ used by an IVF index with N clusters. Let $p = n_{t^*,C}/|C|$ be the in-cluster prevalence of rare token t^* . The centroid is $c = \frac{1}{|C|} \sum_{d \in C} f(d)$, so c contains a term $\alpha_{t^*} p \phi(t^*)$. A query q dominated by t^* has embedding $\approx \alpha_{t^*} \phi(t^*)$ (modulo normalization), and the centroid score it elicits is

$$\langle c, q \rangle \approx \alpha_{t^*}^2 p = \log^2\left(\frac{M}{n_{t^*}}\right) \cdot \frac{n_{t^*,C}}{M/N}. \quad (1)$$

The squared IDF arises because α_{t^*} enters both the centroid (through document weighting) and the query side. This is our *centroid-dilution bound*: as the index uses fewer, larger clusters (small N), $|C|$ grows and the rare-token signal in any one centroid shrinks.

b) Loss condition.: JL projection injects noise of magnitude ϵ into inner products. The rare-token signal is lost when

$$\alpha_{t^*}^2 p \lesssim \epsilon \approx \sqrt{\frac{\log V}{d}}. \quad (2)$$

Increasing d reduces ϵ and so helps rare-token retrieval, but cannot raise $\alpha_{t^*}^2 p$. So dimensionality alone cannot rescue

retrieval when p is too small—i.e., when cluster granularity is too coarse.

B. Indexing Bias

Exact (Flat) nearest-neighbor search over $z = Wf(d)$ preserves these inner products up to JL distortion. ANN structures introduce a *second* source of error, captured by the *indexing bias*

$$\text{Bias}(q, d) = P_{\text{exact}}(\text{ret}(q, d)) - P_{\text{approx}}(\text{ret}(q, d)). \quad (3)$$

For IVF, this manifests as *cluster-selection failure*: the correct cluster is missed when its centroid signal in Eq. (1) falls below the noise floor, even though the document containing t^* would have ranked highly under Flat search of that cluster. For HNSW, it manifests as *sparse connectivity*: documents whose embeddings are dominated by rare tokens occupy low-density regions of the embedding space, and the greedy graph traversal from typical entry points rarely reaches them. Both failure modes attack the same population of queries—those whose relevance hinges on rare, high-IDF tokens—which we call *spear-fishing* queries.

C. Testable Predictions

This framework leads to four experimentally testable predictions, each tied to a specific experiment in Sec. IV:

- 1) **IVF Recall Scaling:** For rare-term and ID queries, IVF Recall@k should increase approximately linearly with probe ratio ($n_{\text{probes}}/n_{\text{lists}}$), but plateau below exact search due to centroid dilution.
- 2) **Exact vs. Approximate Gap:** The performance gap between exact and approximate search will be substantially larger for rare-term and ID queries than for standard semantic queries.
- 3) **Dimensionality Benefit:** Increasing embedding dimensionality d should disproportionately benefit rare-term queries under ANN search by reducing projection noise ϵ .
- 4) **Index Mechanism Differences:** Both IVF and HNSW underperform Flat search on rare queries, but failure patterns might differ, reflecting distinct mechanisms.

IV. EXPERIMENTS

Our experimental design isolates index-induced failures by holding the representation family fixed and varying only the indexing structure. Concretely, we use unsupervised character-trigram embeddings projected to a controlled dimension; this lets us interpret any Flat→ANN gap as an index effect rather than a representation effect.

A. Setup

a) *Datasets and corpus construction.*: We evaluate on two datasets chosen for contrasting query distributions:

- **Home Depot (HD)**: from the public Home Depot product-search relevance corpus. We deduplicate by `product_uid`, yielding 54,667 unique product titles, and use the 11,795 unique customer `search_term`

values with the provided product→query relevance as ground truth.

- **MS MARCO Passage v1.1 (MM)**: we use the validation split (10,047 queries) and flatten each query’s candidate passage list into 82,360 passage documents, marking the passage with `is_selected=1` as relevant.

b) *Query subsets.*: For each dataset we construct three query subsets, each capped at 1,000 queries (random sampling with seed 42 where applicable):

- **Standard**: real user queries with their provided relevance judgments.
- **ID**: a random sample of document identifiers used as queries, with self-relevance ($q=d$). To enable retrieval on these IDs, the identifier string is prepended to each document’s text when indexing this subset. This is the extreme “spear fishing” regime: relevance depends on a single low-frequency token.
- **Rare**: generated queries consisting of unigrams or bigrams (alphabetic, length > 3) that occur in at most 5 documents of the corpus, with their containing documents as relevance set.

c) *Embedding model.*: We use a fully unsupervised pipeline so no learned signal can confound the comparison: `HashingVectorizer` with $2^{18}=262,144$ buckets, character 3-grams (`analyzer='char'`, `across-token mode`—we verified `'char_wb'` gives qualitatively the same trends), followed by a sparse Johnson–Lindenstrauss projection (`SparseRandomProjection`, density 0.9, seed 42) to dimension $d \in \{256, 512, 1024\}$. Embeddings are L2-normalized so that L2 nearest-neighbor search is equivalent to cosine similarity.

d) *Why unsupervised trigram + random projection is the right embedding for this study.*: A natural reviewer question is why we do not use a learned dense encoder (e.g. DPR or Sentence-BERT) as the representation. The answer is methodological: this paper’s claim is about *the index conditional on a fixed representation*, and a supervised encoder *cannot* hold the representation fixed in the way our argument requires. Trigram + JL projection preserves L2 distances up to known multiplicative distortion $\epsilon \propto 1/\sqrt{d}$, so the lexical content of a rare token, an identifier, or a SKU is provably retained in the embedding—which is precisely what defines a spear-fishing query. A supervised semantic encoder is trained to do the opposite: it abstracts away lexical surface form, mapping all five-digit IDs (or all under-represented rare unigrams) to nearly the same point in embedding space because no training pair has ever forced them apart. The Flat baseline on such an encoder would therefore already fail on identifier queries before the index has any chance to bias the result, conflating the two failure modes we are trying to separate. The trigram setup is thus not a baseline of convenience but the correct instrument for isolating the Flat→ANN gap; in the discussion we report a confirmatory experiment with a supervised encoder showing the predicted ceiling-effect collapse on spear-fishing

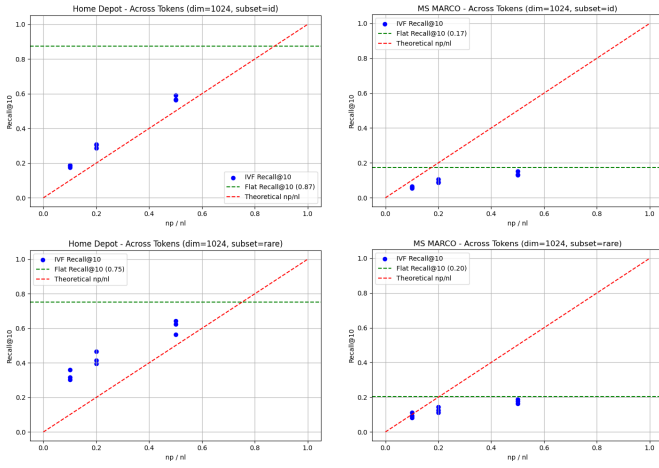


Fig. 1. Recall@10 vs. probe ratio $n_{\text{probes}}/n_{\text{list}}$ for IVF (blue) and Flat (green dashed), $d=1024$. Top: ID queries; bottom: rare queries. Left: Home Depot; right: MS MARCO. The IVF cloud grows with probe ratio but stays well below Flat on every panel.

queries.

e) *Retrieval backends and parameter sweeps.*: All dense methods use FAISS:

- **Flat**: exhaustive L2 search (P_{exact} baseline).
- **IVF**: IndexIVFFlat with $n_{\text{list}} \in \{200, 500, 1000\}$ k-means cells and $n_{\text{probe}} = \lceil n_{\text{list}} \cdot p \rceil$ for $p \in \{0.1, 0.2, 0.5\}$, giving nine $(n_{\text{list}}, n_{\text{probe}})$ configurations per dimension.
- **HNSW**: IndexHNSWFlat with $M=64$, $\text{efConstruction} \in \{100, 200\}$, $\text{efSearch}=100$.
- **BM25**: Anserini/Lucene via Pyserini [16] with default parameters.

f) *Metric and reproducibility.*: We report Recall@10 (we also compute Recall@ $\{1, 5, 20, 50, 100\}$ with consistent trends). All numbers are aggregated over the $\leq 1,000$ queries per subset. The full sweep (over 200 runs per dataset across the cross-product of $\{\text{subset, dimension, index, sweep parameter}\}$) is deterministic given the seed; the released code generates all tables and figures from one JSON results file.

B. IVF Recall Scales with Probe Ratio but Plateaus Below Flat

Figure 1 tests **Prediction 1**. IVF Recall@10 rises monotonically (though sub-linearly in absolute terms) with $n_{\text{probes}}/n_{\text{list}}$, consistent with broader cluster probing finding more correct clusters; but it plateaus well below Flat even at probe-ratio 0.5. The persistent gap supports **Prediction 2**: even when the correct cluster is probed, retrieval within that cluster often fails because the rare-token signal in the centroid was below threshold and the index does not re-rank reliably within the probed cells.

C. Finer Clustering Helps Rare/ID More Than Standard

Figure 2 provides a direct test of the centroid-dilution mechanism. Eq. (1) predicts the per-centroid signal magnitude grows linearly with N (smaller $|C|$ keeps higher in-cluster

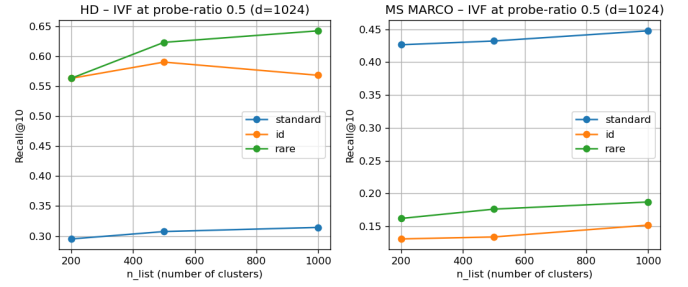


Fig. 2. IVF Recall@10 vs. n_{list} at probe ratio 0.5, $d=1024$. Smaller cluster sizes (larger n_{list}) help *rare* and *ID* queries roughly 2–3 $\times$ more (in relative terms) than standard queries on both datasets, a direct empirical test of the centroid-dilution bound in Eq. (1).

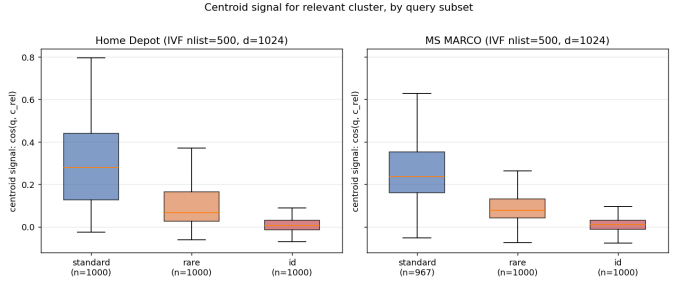


Fig. 3. Per-query centroid signal $\langle q, c_{\text{rel}} \rangle$ measured on the IVF centroid of each query’s relevant cluster, at $n_{\text{list}}=500$, $d=1024$. Boxes show distribution across queries within each subset. The median signal drops $\sim 40\times$ from *standard* to *ID* on Home Depot ($0.279 \rightarrow 0.007$), a direct empirical instantiation of the centroid-dilution bound in Eq. (1).

prevalence p), so spear-fishing queries should benefit more from finer clustering than standard queries. We observe exactly this asymmetry: going from $n_{\text{list}}=200$ to 1000 improves rare-query Recall@10 by $+0.079$ on HD and $+0.025$ on MS MARCO (14% and 15% relative), versus only $+0.019$ and $+0.021$ (6% and 5% relative) for standard queries.

D. Direct Measurement of the Centroid Signal

To move beyond inferring centroid dilution from aggregate Recall@10, we measure it directly per query. For each query q we compute the centroid signal $\langle q, c_{\text{rel}} \rangle$ on the centroid of the cluster containing the relevant document, and the *rank* of that centroid among all 500. Table I summarizes. The median centroid signal collapses monotonically across the three subsets ($0.279 \rightarrow 0.066 \rightarrow 0.007$ on HD)—a $40\times$ drop. The same effect appears from the search side: the relevant cluster is the 2nd-best centroid for the median standard query, but the 216th (out of 500) for the median ID query—indistinguishable from random in the bottom half. At probe-ratio 0.5, 41% of HD-ID queries are *structurally unrecoverable*: their correct cluster is never even probed.

E. Embedding Dimensionality on Exact Search

Increasing d from 256 to 1024 on Flat improves Recall@10 by $+74\%$ relative on HD-ID and $+23\%$ on HD-rare, but only $+16\%$ on HD-standard—consistent with the JL bound

TABLE I

DIRECT CENTROID-DILUTION MEASUREMENT ON THE IVF INDEX ($n_{\text{LIST}}=500$, $d=1024$). HIT@250 = FRACTION OF QUERIES WHOSE RELEVANT CLUSTER RANKS IN THE TOP 250 CENTROIDS AND WOULD THEREFORE BE PROBED AT PROBE-RATIO 0.5.

Dataset	Subset	Med. signal	Med. rank	Hit@250
HD	standard	0.279	2	0.972
HD	rare	0.066	69	0.809
HD	id	0.007	216	0.591
MS MARCO	standard	0.236	20	0.907
MS MARCO	rare	0.077	113	0.794
MS MARCO	id	0.011	236	0.532

TABLE II
RECALL@10 FOR BEST CONFIGURATIONS

Dataset	Subset	Flat	IVF	HNSW	BM25
HD	standard	0.327	0.314	0.321	0.447
HD	id	0.874	0.590	0.327	1.000
HD	rare	0.751	0.642	0.619	0.878
MS MARCO	standard	0.471	0.448	0.446	0.770
MS MARCO	id	0.174	0.152	0.071	0.000 [†]
MS MARCO	rare	0.205	0.187	0.166	0.688

Dense at $d=1024$, across-token trigrams, best ANN sweep.

[†]BM25 cannot tokenize compound passage IDs.

Paired-bootstrap 95% CIs on the Flat-HNSW gap exclude zero for every row; HD-ID: +0.576 [+0.544, +0.607], MM-ID: +0.138 [+0.113, +0.162].

$\epsilon \approx 1/\sqrt{d}$ (rare tokens carry signal closer to the noise floor and gain more from a thicker representation). Critically, this benefit applies to *exact* search only; under HNSW the same $d=1024$ embedding stalls at 0.327 on HD-ID. On MS MARCO, even Flat dense recall is below BM25 on standard and rare subsets—a known limitation of unsupervised character-trigram embeddings on natural-language passages, independent of our index-bias claim, which concerns the Flat→ANN gap rather than the absolute level.

F. Detailed Performance Comparison

Table II validates **Predictions 2–4** at $d=1024$.

Prediction 2 (Exact vs. Approximate gap). The Flat→HNSW gap is order-of-magnitude larger on spear-fishing subsets than on standard queries. On Home Depot, the gap is 0.006 absolute (1.8% relative) for standard queries, 0.132 (17.5% relative) for rare queries, and 0.547 (62.6% relative) for ID queries. On MS MARCO the absolute Recall@10 is lower but the same pattern holds: 0.025 (5.3%) for standard, 0.039 (19.0%) for rare, and 0.103 (59.2%) for ID. The standard-vs-spear-fishing contrast is the central quantitative result of the paper.

Prediction 3 (Dimensionality). Higher embedding dimensionality benefits Flat search on spear-fishing queries far more than on standard queries: HD-ID Flat moves from 0.501 at $d=256$ to 0.874 at $d=1024$ (a 74% relative gain), HD-rare from 0.613 to 0.751 (23%), but HD-standard only from 0.281

to 0.327 (16%). This matches the JL bound $\epsilon \approx \sqrt{\log V/d}$: rare tokens carry signal closer to the noise floor and gain more from reducing ϵ . Crucially, however, dimensionality alone does not close the index gap—HD-ID HNSW remains at 0.327 even at $d=1024$.

Prediction 4 (Index mechanism differences). IVF consistently outperforms HNSW on ID queries (HD: 0.590 vs. 0.327; MM: 0.152 vs. 0.071), but the two are within 0.03 on rare and standard subsets. This is consistent with the theory: ID queries have an extreme connectivity-failure profile that punishes HNSW’s greedy graph traversal more than IVF’s broader cluster probing, while rare-token signals suffer both centroid dilution (IVF) and connectivity loss (HNSW) to comparable degrees.

Honest framing on MS MARCO. BM25 outperforms even Flat dense on MS MARCO standard and rare subsets (0.770 vs. 0.471; 0.688 vs. 0.205). This is a representation limitation of unsupervised trigram embeddings on natural-language passages—not the phenomenon this paper studies. Our claim concerns the Flat→ANN gap, which is well-defined regardless of the absolute level.

Practical visibility (zero-result rate). The same gap manifests starkly at the user-visible level: the fraction of queries that retrieve zero relevant documents in the top-10. On HD-ID, switching from Flat to HNSW takes this from 13% to 71%; on MS MARCO-ID, from 78% to 92%; on standard subsets it barely moves (≤ 2 percentage points on either dataset).

G. The recall-latency trade-off changes shape on spear-fishing queries

The Recall@10 numbers in Table II report HNSW at the FAISS default `efSearch=100`. The complete picture requires sweeping `efSearch` and measuring the latency cost of approaching Flat. Fig. 4 does this for each (dataset, subset) with the standard HNSW recipe $M=16$, $\text{efC}=200$. The HNSW curve has a qualitatively different shape on standard vs. spear-fishing subsets.

a) *Cost-to-close: the corpus-invariant signal.* We summarize the figure by the `efSearch` value required to reach 95% of Flat’s Recall@10, with the resulting latency relative to Flat:

Subset	efS for 95% Flat	ms/q	vs. Flat
HD-standard	100	0.39	$2.7\times$ faster
HD-rare	2,000	6.94	$6.5\times$ slower
HD-ID	8,000	41.9	$40\times$ slower
MM-standard	1,000	3.48	$2.1\times$ slower
MM-rare	16,000	129.6	$79\times$ slower
MM-ID	16,000	132.0	$81\times$ slower

Closing the ID gap to 95% of Flat recall requires 80–160 $\times$ more `efSearch` budget than the standard subset on the same data—an 40–80 $\times$ multiplicative latency factor at our corpus sizes. The corpus-invariant signal is the `efSearch` ratio itself: ANN search cost grows roughly linearly with `efSearch`, so this multiplier carries through any scale of deployment.

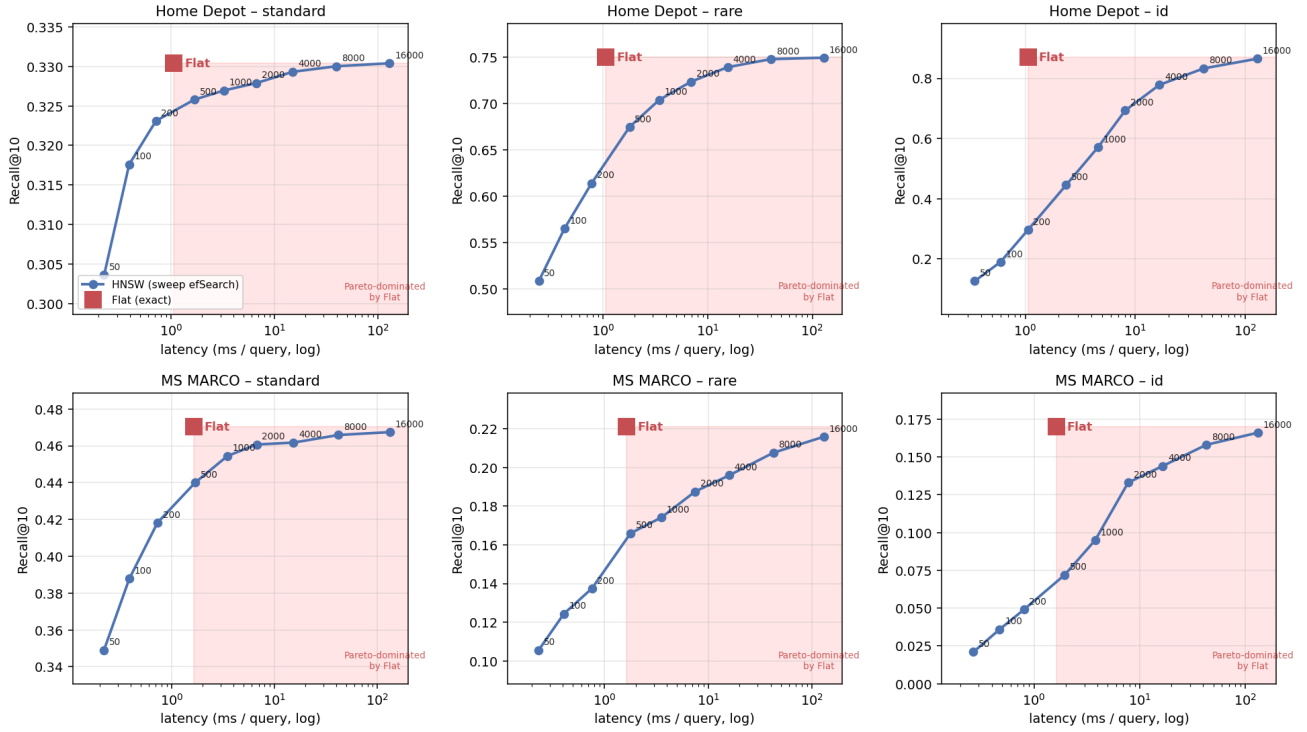


Fig. 4. Recall@10 vs. per-query latency (log-x), single-threaded with warm-up; HNSW configured with the standard FAISS recipe $M=16$, $efC=200$, sweeping $efSearch \in \{50, \dots, 16000\}$ (blue curve, annotated). Flat is one red square. Red shading marks the region strictly dominated by Flat (slower and lower recall). Flat upper-bounds recall by construction. On standard subsets HNSW at $efS=50$ –100 is 2–5 $\times$ faster than Flat at $\geq 92\%$ of Flat’s recall—the canonical ANN sweet spot. On ID subsets the HNSW curve climbs slowly: HD-ID requires $efS=8000$ (40 $\times$ Flat’s latency) to reach 95% of Flat’s recall; MM-ID needs $efS=16000$ (81 $\times$).

b) Standard queries: the canonical ANN sweet spot.: On HD-standard HNSW $efS=100$ delivers 96% of Flat’s recall at 0.39 ms/q—2.7 $\times$ faster than Flat’s 1.07 ms/q. On MM-standard the same operating point delivers 82% recall at 0.39 ms/q (4.2 $\times$ faster). At $efS=50$, HD-standard achieves 4.7 $\times$ speedup at 92% recall. This is exactly what HNSW is designed for and used in production.

c) Corpus-size caveat.: At our $N \approx 10^5$, FAISS Flat is fast (~ 1 –1.6 ms/q) because dense matrix–vector products parallelize well across SIMD; HNSW’s logarithmic-speedup advantage is modest. At $N=10^7$ Flat would slow linearly (~ 100 ms/q) while HNSW $efS=100$ grows logarithmically, so HNSW would clearly beat Flat in latency on *standard* queries. The *cost-to-close ratio* is what generalizes: at any scale, reaching Flat-level recall on ID requires 80–160 $\times$ the $efSearch$ budget of standard, and therefore consumes a substantial fraction of HNSW’s speedup advantage.

H. Supervised encoders: confirmation of the methodological argument

The Setup paragraph argued, on theoretical grounds, that a supervised semantic encoder cannot serve as a controlled representation for studying index bias because it abstracts away the lexical surface form that defines a spear-fishing query. We tested this directly by re-running the full grid on all-MiniLM-L6-v2 (Sentence-BERT, 384-dim). On MM-

standard the encoder nearly *doubles* Flat recall (0.471 $\rightarrow$ 0.894)—exactly the regime it was trained for. On HD-ID, Flat recall *collapses* from 0.874 (trigram) to 0.010 (MiniLM); on MM-ID, from 0.174 to 0.001. The Flat $\rightarrow$ HNSW gap is meaningless on these subsets under MiniLM because Flat itself returns essentially nothing relevant. This is the predicted ceiling effect and confirms that the trigram setup, far from being a baseline of convenience, is the necessary instrument for isolating the index-stage failure.

V. DISCUSSION AND CONCLUSION

We formalized *indexing bias* and the *centroid-dilution bound*, predicted four observable consequences, and confirmed each: IVF recall scales sub-linearly with probe ratio and plateaus below Flat; finer clustering helps rare/ID disproportionately; the per-query centroid signal collapses $\sim 40\times$ from standard to ID queries (Table I); the $efSearch$ budget required to reach 95% of Flat recall is 80–160 $\times$ larger on ID than on standard queries (Fig. 4), so closing the spear-fishing gap consumes a substantial fraction of HNSW’s speed advantage. The picture reframes the sparse-vs-dense debate as a joint choice of representation *and* indexing structure: a clean standard-query trade-off from HNSW is paid for by a steep cost-to-close on the spear-fishing population. Two concrete avenues follow: *query-type-aware routing* that diverts spear-fishing traffic to BM25 or Flat, and *index-aware training* that

penalizes placing rare-feature embeddings in poorly reachable regions. Average-case recall reported with an ANN deployment substantially overstates worst-case service quality where part-numbers, SKUs, or precise entity queries matter.

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